On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of those principles will lead to a zero grade and is subject to harsh discipline issues.

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1) $A=\begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -2\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & -1 & 1 & 0 \\ -1 & 0 & 1 & 1 \end{bmatrix}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$

a) $-8R_1+R_2 \to R_2$

$3\alpha-8=0 \quad \boxed{\alpha=\frac{8}{3}}$

$\Rightarrow \begin{bmatrix} 1 & \frac{8}{3} & -\gamma & \frac{8}{3} \\ 0 & 5-\frac{64}{3} & -\gamma & 1-\frac{64}{3} \\ 0 & \beta-\frac{16}{3} & 3+2\gamma & \frac{64}{9}-\frac{16}{3} \\ 0 & 14-\frac{8}{3} & -6+\gamma & 2-\frac{8}{3} \end{bmatrix} \begin{matrix} \\ \\ \to \frac{16}{9} \\ \to -\frac{2}{3} \end{matrix}$

$m+2=0$

$\boxed{m=-2}$

$-R_2+R_3 \to R_3 \Rightarrow \frac{49}{3}+\beta-\frac{16}{3}=0 \Rightarrow \boxed{\beta=-11}$

$\boxed{k=-2}$

$\begin{bmatrix} 1 & \frac{8}{3} & -\gamma & \frac{8}{3} \\ 0 & \frac{49}{3} & -\gamma & -\frac{61}{3} \\ 0 & 0 & 3+3\gamma & \frac{16}{9} \\ 0 & 0 & -3\gamma & \frac{26}{9} \end{bmatrix}$

$-\frac{1}{2}(3+3\gamma)-3\gamma=0$

$\boxed{\gamma=1}$

b-) $U=\begin{bmatrix} 1 & \frac{8}{3} & -1 & 8/3 \\ 0 & \frac{49}{3} & -1 & -61/3 \\ 0 & 0 & 6 & 16/9 \\ 0 & 0 & 0 & 26/18 \end{bmatrix}$

c) $E_{43}E_{32}E_{21} \cdot A = U$

$\begin{bmatrix} 1 & 0 \\ -8 & 1 \\ m & 0 \\ -1 & 0 \end{bmatrix}$

3.) $H = I - 2uu^T$

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \begin{bmatrix} u_1 & u_2 & \cdots & u_n \end{bmatrix}$$

a-> $(I - 2uu^T).(I - 2uu^T) = I^2 - 2uu^T - I.2uu^T + 4uu^Tuu^T$

=> $I - 2uu^T - 2uu^T + 4\,u\underbrace{u^Tu}_{I}u^T$ => $I - 4uu^T + 4uu^T$ => $\boxed{H^2 = I}$

4-) $V(0) = 10$ liters
$W(0) = 40$ liters

a-) $\frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} v \\ w \end{bmatrix}$

b) $|A - \lambda I| = \begin{vmatrix} 2-\lambda & -2 \\ -1 & 1-\lambda \end{vmatrix}$

$(2-\lambda)(1-\lambda) - 2 = 0$

$2 - 2\lambda - \lambda + \lambda^2 - 2 = 0$

$\lambda^2 - 3\lambda = 0$

$\lambda_1 = 0$ $\qquad \lambda_1 + \lambda_2 = 3$ Trace

dependent column

$\lambda_2 = 3$

$\lambda_1 = 0 \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \Rightarrow \quad x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda_2 = 3 \begin{vmatrix} 2-3 & -2 \\ -1 & 1-3 \end{vmatrix} = \begin{vmatrix} -1 & -2 \\ -1 & -2 \end{vmatrix} \Rightarrow x_2 = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$

$U(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{\lambda_1 t} \cdot c_1 + \begin{bmatrix} -2 \\ 1 \end{bmatrix} e^{\lambda_2 t} \cdot c_2$

$U(0) = \begin{bmatrix} 10 \\ 40 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot c_1 + \begin{bmatrix} -2 \\ 1 \end{bmatrix} c_2$

$10 = c_1 - 2c_2$
$40 = c_1 + c_2$

$c_1 = 30$
$c_2 = 10$

$U(t) = \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^{0.t} \cdot 30 + \begin{bmatrix} -2 \\ 1 \end{bmatrix} \cdot e^{3.t} \cdot 10$

$U(ss)$ $\quad$!
$t \to \infty \quad = \quad K \qquad e^{\lambda_2 t} \to \infty \quad \Rightarrow$ Two tank will equal